

All India Test Series

NEET (2025-26)

**DURATION: 180 Minutes****DROPPER****M. MARKS: 720**DATE: 07/12/2025**ANSWER KEY**

1. (2)	46. (3)	91. (2)	136. (1)
2. (2)	47. (2)	92. (4)	137. (2)
3. (2)	48. (4)	93. (3)	138. (3)
4. (4)	49. (4)	94. (4)	139. (1)
5. (1)	50. (2)	95. (2)	140. (2)
6. (3)	51. (3)	96. (3)	141. (3)
7. (2)	52. (2)	97. (3)	142. (2)
8. (2)	53. (4)	98. (2)	143. (2)
9. (3)	54. (1)	99. (2)	144. (1)
10. (4)	55. (3)	100. (3)	145. (1)
11. (1)	56. (2)	101. (3)	146. (3)
12. (2)	57. (4)	102. (1)	147. (2)
13. (1)	58. (2)	103. (3)	148. (3)
14. (2)	59. (4)	104. (3)	149. (4)
15. (1)	60. (2)	105. (2)	150. (3)
16. (2)	61. (2)	106. (2)	151. (3)
17. (1)	62. (3)	107. (1)	152. (2)
18. (3)	63. (2)	108. (1)	153. (3)
19. (1)	64. (1)	109. (2)	154. (2)
20. (4)	65. (1)	110. (3)	155. (4)
21. (3)	66. (3)	111. (1)	156. (3)
22. (4)	67. (3)	112. (2)	157. (4)
23. (1)	68. (3)	113. (3)	158. (3)
24. (2)	69. (1)	114. (4)	159. (1)
25. (3)	70. (1)	115. (2)	160. (4)
26. (3)	71. (1)	116. (2)	161. (4)
27. (3)	72. (2)	117. (1)	162. (2)
28. (3)	73. (1)	118. (2)	163. (4)
29. (1)	74. (2)	119. (2)	164. (2)
30. (2)	75. (2)	120. (4)	165. (4)
31. (1)	76. (3)	121. (2)	166. (3)
32. (1)	77. (2)	122. (1)	167. (1)
33. (4)	78. (4)	123. (3)	168. (2)
34. (2)	79. (3)	124. (1)	169. (4)
35. (2)	80. (4)	125. (2)	170. (2)
36. (3)	81. (1)	126. (3)	171. (2)
37. (1)	82. (1)	127. (2)	172. (3)
38. (1)	83. (1)	128. (4)	173. (1)
39. (1)	84. (2)	129. (3)	174. (2)
40. (2)	85. (1)	130. (1)	175. (1)
41. (2)	86. (1)	131. (1)	176. (1)
42. (3)	87. (2)	132. (4)	177. (2)
43. (4)	88. (2)	133. (4)	178. (1)
44. (1)	89. (2)	134. (2)	179. (1)
45. (1)	90. (3)	135. (3)	180. (3)

Hints & Solutions

1. (2)

Due to the symmetry of the square sheet, the moment of inertia about any axis passing through its center of mass and lying in its plane is constant. This value is always $I = Ma^2/12$, regardless of the angle θ . Therefore, the graph of I versus θ is a horizontal line.

(NCERT Ed. 2025-26, 11th, Page No. 111)

2. (2)

The gravitational field E at the center of a semicircular wire of mass M and length L is given by:

$$E = 2\pi GM/L^2$$

Rearranging for mass M :

$$M = (E \times L^2) / (2\pi G)$$

Substituting the given values:

$$M = (4\pi G \text{ N/kg} \times (1 \text{ m})^2) / (2\pi G) = 2 \text{ kg}$$

(NCERT Ed. 2025-26, 11th, Page No. 129)

3. (2)

Young's Modulus (also known as the modulus of elasticity) is an intrinsic property of a material. the Young's Modulus remains a constant value for that material.

(NCERT Ed. 2025-26, 11th, Page No. 117)

4. (4)

Principle: Torque $\tau = \text{Force} \times \text{Perpendicular Lever Arm}$.

Force: The force is the weight (mg), which acts vertically downwards.

$$\tan(\theta) = \frac{h}{\text{base}}$$

$$\text{Therefore, base} = \frac{h}{\tan(\theta)} = h \cot(\theta).$$

Calculate Torque:

$$\tau = (mg) \times (h \cot(\theta)) = mgh \cot(\theta).$$

(NCERT Ed. 2025-26, 11th, Page No. 96)

5. (1)

Force on m_1 from m_2 (along +x axis):

$$F_{12} = G \frac{m_1 m_2}{r^2} \hat{i} = G \frac{2 \times 9}{3^2} \hat{i} = 2G \hat{i}$$

Force on m_1 from m_3 (along +y axis):

$$F_{13} = G \frac{m_1 m_3}{r^2} \hat{j} = G \frac{2 \times 32}{4^2} \hat{j} = 4G \hat{j}$$

Net Force on m_1 :

$$F_{\text{net}} = F_{12} + F_{13} = 2G \hat{i} + 4G \hat{j}$$

Initial Acceleration ($a = F_{\text{net}} / m_1$):

$$a = \frac{2G \hat{i} + 4G \hat{j}}{2} = G(\hat{i} + 2\hat{j}) \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

6. (3)

$$F = \frac{GM}{(3R)^2} \times m = \frac{GMm}{9R^2}$$

(NCERT Ed. 2025-26, 11th, Page No. 122)

7. (2)

$$r = r_1 - r_2$$

$$r = (3\hat{i} + \hat{j} + 2\hat{k}) - (2\hat{i} - \hat{j} + \hat{k})$$

$$r = (1\hat{i} + 2\hat{j} + 1\hat{k}) \text{ m}$$

Calculate the torque (τ) using the cross-product formula $\tau = r \times F$:

$$\tau = (\hat{i} + 2\hat{j} + \hat{k}) \times (\hat{i} + 2\hat{j} - 3\hat{k})$$

$$\tau = \hat{i}[(2)(-3) - (1)(2)] - \hat{j}[(1)(-3) - (1)(1)]$$

$$+ \hat{k}[(1)(2) - (2)(1)]$$

$$\tau = -8\hat{i} - \hat{j}(-4) + 0\hat{k}$$

$$\tau = (-8\hat{i} + 4\hat{j}) \text{ Nm}$$

(NCERT Ed. 2025-26, 11th, Page No. 121)

8. (2)

For perfectly rigid body, strain = 0

$$\Rightarrow Y = \frac{\text{stress}}{\text{strain}} = \text{infinite}$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

9. (3)

$$g_p = \frac{GM_p}{R_p^2}$$

$$g_{\text{new}} = \frac{G(2M_p)}{\left(\frac{R_p}{2}\right)^2}$$

$$g_{\text{new}} = \frac{G(2M_p)}{\frac{R_p^2}{4}}$$

$$g_{\text{new}} = 8 \times \left(\frac{GM_p}{R_p^2} \right)$$

$$g_{\text{new}} = 8 \times g_p$$

(NCERT Ed. 2025-26, 11th, Page No. 105)

10. (4)

From Young's Modulus, $F = Y \times A \times \left(\frac{\Delta l}{l} \right)$.

Since Volume (V) is constant, $V = A \times l$, so $1 = \frac{V}{A}$.

$$\text{Substitute 1: } F = Y \times A \times \left(\frac{\Delta l}{\frac{V}{A}} \right) = \left(\frac{Y \times \Delta l}{V} \right) \times A^2$$

Since $Y, \Delta l$, and V are the same for both scenarios, F is proportional to A^2 .

$$F_p / F_q = (A_p / A_q)^2$$

$$F_p / F = \left(\frac{A}{4A} \right)^2 = \left(\frac{1}{4} \right)^2 = \frac{1}{16}$$

$$F_p = \frac{F}{16}$$

(NCERT Ed. 2025-26, 11th, Page No. 115)

11. (1)

According to work energy theorem,

$$W = \Delta KE = 0 - \left(\frac{1}{2} I \omega^2 \right)$$

$$W = \frac{1}{2} \times \frac{50 \times (1)^2}{2} \times 16$$

$$= 200 \text{ J}$$

(NCERT Ed. 2025-26, 11th, Page No. 106)

12. (2)

Stress: Since the wires are connected in series, they are subjected to the same tensile force (F). the stress is the same in both.

Strain: the stress is the same but the Young's Moduli are different ($Y_{\text{brass}} \neq Y_{\text{alum}}$), the strain will be different in each wire.

(NCERT Ed. 2025-26, 11th, Page No. 119)

13. (1)

$$\alpha = 6t^2 - 2t$$

$$\int_{10}^{\omega} d\omega = \int_0^5 (6t^2 - 2t) dt$$

$$\omega - 10 = \left(\frac{6t^3}{3} - \frac{2t^2}{2} \right)_0^5$$

$$\omega - 10 = (2t^3 - t^2)_0^5$$

$$\omega = 235 \text{ rad/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 135)

14. (2)

Elastic forces are not always conservative.

Imagine plotting the Force vs. Extension graph for a real rubber band: As you pull the rubber band, you follow an upper curve on the graph. The work you do is the area under this curve. As you let the rubber band contract, it follows a lower curve on the graph. The work the band does on you is the area under this lower curve.

Because the work done on the material is greater than the work done by the material when it returns, the net work over a closed loop is not zero.

(NCERT Ed. 2025-26, 11th, Page No. 106)

15. (1)

Initial Potential Energy (on the surface):

$$U_i = -\frac{GMm}{R}$$

Final Potential Energy (at height $h = R/2$):

$$U_f = -\frac{GMm}{R + \frac{R}{2}} = -\frac{GMm}{\frac{3R}{2}} = -\frac{2}{3} \frac{GMm}{R}$$

Change in Potential Energy (ΔU)

$$= U_f - U_i$$

$$\Delta U = \left(-\frac{2}{3} \frac{GMm}{R} \right) - \left(-\frac{GMm}{R} \right)$$

$$\Delta U = \frac{GMm}{R} \left(1 - \frac{2}{3} \right) = \frac{1}{3} \frac{GMm}{R}$$

Since $g = \frac{GM}{R^2}$, we can write $GM = gR^2$.

$$\Delta U = \frac{1}{3} \frac{gR^2 m}{R} = \frac{1}{3} mgR$$

(NCERT Ed. 2025-26, 11th, Page No. 106)

16. (2)

Gravitational force is a conservative force.

$$\Rightarrow W_1 = W_2 = W_3$$

(NCERT Ed. 2025-26, 11th, Page No. 107)

17. (1)

$$\text{Shear stress} = \tau = \frac{F \sin \theta}{A / \cos \theta}$$

$$= \left(\frac{F}{2A} \right) \times \sin(2\theta)$$

Therefore, for maximum shear stress:

$$2\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{4}$$

(NCERT Ed. 2025-26, 11th, Page No. 115)

18. (3)

As disc is lying in the x - z plane, so applying perpendicular axis theorem:

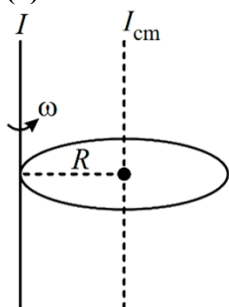
$$I_x + I_z = I_y$$

$$30 + I_z = 40$$

$$\Rightarrow I_z = 40 - 30 = 10 \text{ kg m}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 119)

19. (1)



$$I = I_{cm} + MR^2$$

$$= (MR^2) + MR^2$$

$$I = 2MR^2; \omega = 2\pi n$$

$$KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \times (2MR^2) \times (2\pi n)^2$$

$$\text{or } KE = 4\pi^2 n^2 MR^2$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

20. (4)

$$\vec{E} = \frac{\vec{F}}{m} = \frac{6\hat{i}}{30 \times 10^{-3}} = (200\hat{i}) \text{ N/kg}$$

(NCERT Ed. 2025-26, 11th, Page No. 114)

21. (3)

$$\frac{E_2}{E_1} = \left(\frac{r_1}{r_2} \right)^2$$

$$\frac{E_2}{8 \text{ N/kg}} = \left(\frac{10 \text{ cm}}{15 \text{ cm}} \right)^2$$

$$\frac{E_2}{8} = \left(\frac{2}{3} \right)^2$$

$$\frac{E_2}{8} = \frac{4}{9}$$

$$E_2 = 8 \times \frac{4}{9} = \frac{32}{9} \text{ N/kg}$$

(NCERT Ed. 2025-26, 11th, Page No. 133)

22. (4)

Since, $I_z = I_x + I_y$

$$= mr_x^2 + mr_y^2$$

$$= m(r_x^2 + r_y^2)$$

For particle 1, $= m(3^2 + 2^2) = 26 \text{ unit}$

For particle 2, $m[1^2 + (-1)^2] = 2 \times (2)$

$$= 4 \text{ unit}$$

For particle 3, $m[1^2 + 1^2] = 2 \times (2)$

$$= 4 \text{ unit}$$

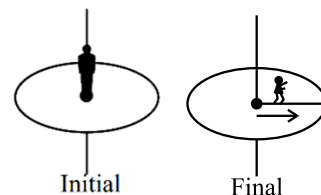
For particle 4, $m(0) = 0$

Total inertia $= 26 + 4 + 4 = 34 \text{ units}$

(NCERT Ed. 2025-26, 11th, Page No. 110)

23. (1)

Applying conservation of angular momentum:



$$I_1 \omega_1 = I_2 \omega_2$$

$$\Rightarrow I_1 \times \frac{2\pi}{T_1} = I_2 \omega_2$$

$$\Rightarrow 100 \times \frac{2\pi}{10} = [100 + 50 \times (2)^2] \times \omega_2$$

on solving, $\omega_2 = \frac{2\pi}{30} \text{ rad/sec}$

(NCERT Ed. 2025-26, 11th, Page No. 120)

24. (2)

$$\Delta l = \frac{F \times l}{Y \times A}$$

Since the volume (V) is fixed, we have $V = A \times l$,

which means $A = \frac{V}{l}$.

$$\Delta l = \frac{F \times l}{Y \times \left(\frac{V}{l} \right)}$$

$$\Delta l = \frac{F}{Y \times V} \times l^2$$

Since F , Y and V are constants, Δl is directly proportional to l^2 ($\Delta l \propto l^2$).

Therefore, the graph of Δl versus l^2 is a straight line.

(NCERT Ed. 2025-26, 11th, Page No. 116)

25. (3)

$$\frac{1}{2} I \omega^2 = 10$$

$$\Rightarrow \frac{5}{2} \omega^2 = 10 \Rightarrow \omega = 2 \text{ rad/sec}$$

Angular Momentum

$$L = I \omega = 5 \times 2 = 10 \text{ joule-sec.}$$

(NCERT Ed. 2025-26, 11th, Page No. 122)

26. (3)

$$F = \frac{Gm(5x - m)}{d^2}$$

$$\frac{dF}{dm} = \frac{G}{d^2}(5x - 2m) = 0$$

$$\Rightarrow m = 2.5x$$

$$\Rightarrow \text{both masses should be } 2.5x$$

$$\Rightarrow \text{mass transferred} = \frac{x}{2}$$

(NCERT Ed. 2025-26, 11th, Page No. 131)

27. (3)

Total elongation = Elongation due to self-weight + Elongation due to block

$$\Delta L = \frac{MgL}{2AY} + \frac{mgL}{AY}$$

(NCERT Ed. 2025-26, 11th, Page No. 111)

28. (3)

In case A,

$$\Delta l = \frac{50 \times L}{AY} = 2 \text{ mm}$$

In case B

$$\Delta l' = \Delta l_1 + \Delta l_2$$

$$= \frac{100 \times \frac{L}{2}}{AY} + \frac{100 \times \frac{L}{2}}{AY}$$

$$\Delta l' = \frac{100L}{AY}$$

$$\Delta l' = 2\Delta l = 4 \text{ mm}$$

(NCERT Ed. 2025-26, 11th, Page No. 104)

29. (1)

$$L = mvr_{\perp} = mvH$$

$$= 2 \times (u \cos 60^\circ) \times \frac{u^2 \sin^2 60^\circ}{2g}$$

$$= \frac{20 \times 400 \times 3}{4 \times 20}$$

$$= 300 \text{ kg m}^2/\text{s}$$

(NCERT Ed. 2025-26, 11th, Page No. 116)

30. (2)

The moment of inertia is minimized when the axis of rotation passes through the system's center of mass (CM).

Let m be at the origin. The CM is at a distance

$$x_{cm} = \frac{m(0) + 2m(r)}{m + 2m} = \frac{2r}{3} \text{ from mass } m.$$

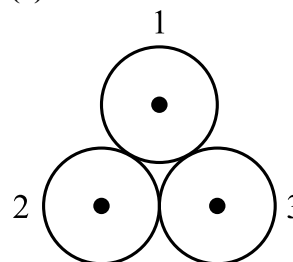
$$I_{\min} = m\left(\frac{2r}{3}\right)^2 + 2m\left(\frac{r}{3}\right)^2$$

$$I_{\min} = m\left(\frac{4r^2}{9}\right) + 2m\left(\frac{r^2}{9}\right) = \frac{4mr^2 + 2mr^2}{9}$$

$$= \frac{6mr^2}{9} = \frac{2}{3}mr^2$$

(NCERT Ed. 2025-26, 11th, Page No. 107)

31. (1)



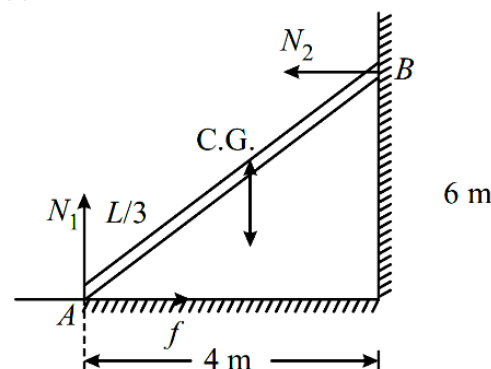
$$PE = PE_{12} + PE_{23} + PE_{31}$$

$$= -\frac{GM^2}{2R} - \frac{GM^2}{2R} - \frac{GM^2}{2R}$$

$$\Rightarrow PE = -\frac{3GM^2}{2R}$$

(NCERT Ed. 2025-26, 11th, Page No. 130)

32. (1)



Translational at equilibrium:

$$\Sigma \vec{F} = 0 \quad \text{i.e.,} \quad \Sigma F_x = 0, \quad \Sigma F_y = 0$$

$$\text{So, } N_2 = f$$

$$N_1 = Mg$$

At rotational equilibrium:

$$\Rightarrow (Mg \cos \theta) \cdot \frac{L}{3} = (N_2 \sin \theta)L$$

$$\Rightarrow N_2 = \frac{Mg}{3} \cdot \cot \theta$$

$$\text{or } N_2 = \frac{500}{3} \times \frac{4}{6}$$

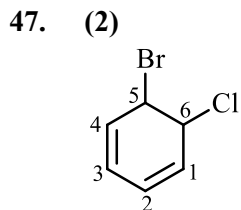
$$\Rightarrow N_2 \approx 111 \text{ N}$$

(NCERT Ed. 2025-26, 11th, Page No. 121)

33. (4)
As all the forces passed through point A .
So, τ_{Net} due to all forces = 0 Nm
(NCERT Ed. 2025-26, 11th, Page No. 134)
34. (2)
Within elastic limit,
 Y = slope of stress-strain graph
(NCERT Ed. 2025-26, 11th, Page No. 106)
35. (2)
 $F \times R = I\alpha \Rightarrow F \times 0.10 = 0.40 \times \alpha$
 $\alpha = 0.25F \rightarrow 100 = 0 + (0.25F) \times 10 \Rightarrow F = 40 \text{ N}$
(NCERT Ed. 2025-26, 11th, Page No. 120)
36. (3)
 $g' = g - R\omega^2 \cos^2 \theta$
At poles $\theta = 90^\circ$,
hence $g' = g$ for any value of ω .
(NCERT Ed. 2025-26, 11th, Page No. 111)
37. (1)
Statement 1 is correct. This is because even though the position vector changes, the perpendicular distance from the point to the line of motion remains constant, and the momentum is constant, resulting in a constant angular momentum.
Statement 2 is correct. The torque produced by a couple is independent of the point about which it is calculated.
Statement 3 is correct. This is the principle of conservation of angular momentum.
Statement 4 is incorrect. The perpendicular axis theorem applies to planar bodies (2D objects). The axes must be concurrent, but they do not necessarily have to pass through the centre of mass. Therefore, statements 1, 2, and 3 are correct.
(NCERT Ed. 2025-26, 11th, Page No. 122)
38. (1)
For the rod to be in rotational equilibrium, the total counter-clockwise torque must equal the total clockwise torque about the pivot.
Counter-clockwise torque (τ_{ccw}) = Force $\times$ Distance
= $(8.0 \sin(\phi)) \times 25 \text{ cm}$
Clockwise torque (τ_{cw}) = Force $\times$ Distance
= $(10 \sin(\theta)) \times 75 \text{ cm}$
 $(8.0 \sin(\phi)) \times 25 = (10 \sin(\theta)) \times 75$
 $200 \sin(\phi) = 750 \sin(\theta)$
 $\sin(\phi) = (750 / 200) \sin(\theta)$
 $\sin(\phi) = 3.75 \sin(\theta)$
(NCERT Ed. 2025-26, 11th, Page No. 120)

39. (1)
Elastic potential energy
 $= \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$
 $= \frac{1}{2} \times \frac{30}{A} \times \frac{0.02}{8} \times Al$
 $\Rightarrow 0.3 \text{ J}$
(NCERT Ed. 2025-26, 11th, Page No. 108)
40. (2)
Gravitational force is conservative force. It is independent of medium, temperature and pressure. It is always attractive in nature.
(NCERT Ed. 2025-26, 11th, Page No. 117)
41. (2)
Bulk modulus of incompressible liquid is infinite. Volume of incompressible liquid does not change by applying force
(NCERT Ed. 2025-26, 11th, Page No. 122)
42. (3)
 $\vec{L} = \vec{r} \times \vec{p}$
 $\Rightarrow |L| = rp \sin \theta = 1 \times 2 \times 6\sqrt{3} \times \sin 60^\circ$
 $= 6\sqrt{3} \times \sqrt{3}$
 $= 18 \text{ kgm}^2/\text{s}$
(NCERT Ed. 2025-26, 11th, Page No. 122)
43. (4)
 $\frac{\Delta V}{V} = 3 \frac{\Delta a}{a} = 3 \times (0.01)$
(NCERT Ed. 2025-26, 11th, Page No. 117)
44. (1)
 $I\omega + I(2\omega) = 2I\omega_f$
 $\Rightarrow 3I\omega = 2I\omega_f$
 $\therefore \frac{\omega_f}{\omega} = \frac{3}{2}$
(NCERT Ed. 2025-26, 11th, Page No. 106)
45. (1)
Gravitational field due to thin uniform circular ring of mass M and radius R on its axis at distance x from its centre is equal to $\frac{GMx}{(x^2 + R^2)^{\frac{3}{2}}}$ and it is maximum at $x = \frac{R}{\sqrt{2}}$.
(NCERT Ed. 2025-26, 11th, Page No. 116)

46. (3)
Work is path function.
(NCERT Ed. 2025-26 11th Part-I Page No. 138)



5-Bromo-6-chlorocyclohexa-1,3-diene
(NCERT Ed. 2025-26 11th Part-II Page No. 266)

48. (4)
$$w = -2.303 RT \log \left(\frac{P_1}{P_2} \right)$$

$$= -2.303 \times 8.314 \times 300 \times \log \left(\frac{20}{2} \right)$$

$$= -5744.14 \text{ J} = -5.74 \text{ kJ}$$

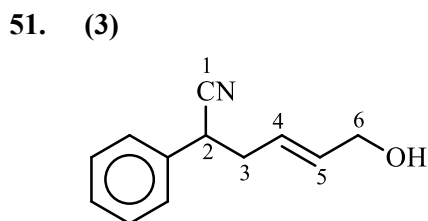
(NCERT Ed. 2025-26 11th Part-I Page No. 148)

49. (4)

Esters	Alkoxycarbonyl
Carboxylic acids	Carboxy
Ethers	Alkoxy
Aldehydes	Formyl

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

50. (2)
 $\Delta U = q + w$
 $= 20 - 1(40 - 10) \times 101.3 \times 10^{-3} = 16.96 \text{ kJ}$
(NCERT Ed. 2025-26 11th Part-I Page No. 140)



6-Hydroxy-2-phenylhex-4-en-1-nitrile
(NCERT Ed. 2025-26 11th Part-II Page No. 266)

52. (2)
On applying eq (1) + 2 × eq (2)
 $A \rightarrow 2C + D$
 $\therefore \Delta H = +80 + 2(-60) = -40 \text{ kJ mol}^{-1}$
 $\therefore 2C + D \rightarrow A, \Delta H = +40 \text{ kJ mol}^{-1}$
(NCERT Ed. 2025-26 11th Part-I Page No. 152)

53. (4)
 $C_6H_5OCH_3$: Anisole
: phenyl methyl ether
: methoxy benzene
(NCERT Ed. 2025-26 11th Part-II Page No. 262)

54. (1)
• $\Delta_f H^\circ$ of an element is zero in its reference state
• Reference state of bromine is $Br_2(l)$.
(NCERT Ed. 2025-26 11th Part-I Page No. 149)

55. (3)
-
- 2,5-dimethyl-6-oxohept-3-yn-1-oyl chloride
(NCERT Ed. 2025-26 11th Part-II Page No. 266)

56. (2)
Those properties which depends on the quantity or size of matter present are known as extensive properties. For example: entropy
(NCERT Ed. 2025-26 11th Part-I Page No. 144)

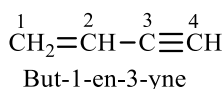
57. (4)
Priority order: $-\text{CONH}_2 > -\text{CHO} > -\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-$
(NCERT Ed. 2025-26 11th Part-II Page No. 266)

58. (2)
Enthalpy of atomization is the enthalpy change when one mole of compound is completely converted into gaseous atoms.
(NCERT Ed. 2025-26 11th Part-I Page No. 153)

59. (4)
-
- : 5-Bromo-2-hydroxybenzonitrile.
(NCERT Ed. 2025-26 11th Part-II Page No. 269)

60. (2)
For spontaneous reaction:
 $\Delta G < 0$
 $\Delta H - T\Delta S < 0$
 $T > \frac{\Delta H}{\Delta S} = \frac{200}{0.4} = 500 \text{ K} = 227^\circ\text{C}$
(NCERT Ed. 2025-26 11th Part-I Page No. 161)

61. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 268)

62. (3)

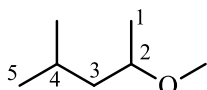
For insulated vessel, $q = 0$

Since work is done on the system, so $w > 0$

Now, $\Delta U = q + w = w \quad \therefore \Delta U > 0$

(NCERT Ed. 2025-26 11th Part-I Page No. 142)

63. (2)



2-Methoxy-4-methylpentane

(NCERT Ed. 2025-26 11th Part-II Page No. 266)

64. (1)

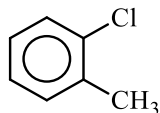
$$\Delta H = \Delta U + \Delta n_g RT$$

$$= 15 - 2 \times 2 \times 10^{-3} \times 300$$

$$= 13.8 \text{ kcal mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 143)

65. (1)



: 1-Chloro-2-methylbenzene
: 2-Chlorotoluene

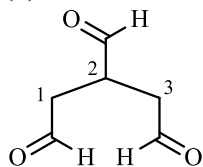
(NCERT Ed. 2025-26 11th Part-II Page No. 269)

66. (3)

No work is done during free expansion of an ideal gas during a irreversible process because expansion of a gas in vacuum ($P_{\text{ex}} = 0$) is called free expansion.

(NCERT Ed. 2025-26 11th Part-I Page No. 142)

67. (3)



Propane-1,2,3-tricarbaldehyde

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

68. (3)

$$\Delta_r H = \Sigma(\text{BE})_R - \Sigma(\text{BE})_P$$

$$= (4\text{BE}_{\text{C-H}} + \text{BE}_{\text{C=C}} + \text{BE}_{\text{H-H}}) - (\text{BE}_{\text{C-C}} + 6\text{BE}_{\text{C-H}})$$

$$= (4 \times 414 + 615 + 435) - (615 + 6 \times 414)$$

$$= -125 \text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 154)

69. (1)

Tetrahydrofuran	
Tropone	
Thiophene	
Naphthalene	

(NCERT Ed. 2025-26 11th Part-II Page No. 262)

70. (1)

Isothermal process	$\Delta U = 0$
Adiabatic process	$q = 0$
Isobaric process	$\Delta P = 0$
Isochoric process	$w = 0$

(NCERT Ed. 2025-26 11th Part-I Page No. 142)

71. (1)

A series of organic compounds each containing a characteristic functional group forms a homologous series

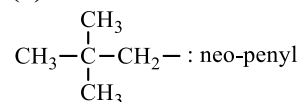
(NCERT Ed. 2025-26 11th Part-II Page No. 262)

72. (2)

When gaseous mole increases, entropy of process increases.

(NCERT Ed. 2025-26 11th Part-I Page No. 158)

73. (1)



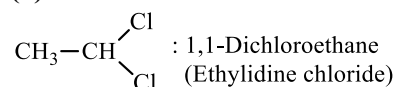
(NCERT Ed. 2025-26 11th Part-II Page No. 263)

74. (2)

The entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero. This is called third law of thermodynamics.

(NCERT Ed. 2025-26 11th Part-I Page No. 161)

75. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 266)

76. (3)

$\Delta_f H$ is defined when one mole of a substance is formed from its elements in their standard states.

(NCERT Ed. 2025-26 11th Part-I Page No. 149)

77. (2)

Catechol	Benzene-1,2-diol
Resorcinol	Benzene-1,3-diol
Hydroquinone	Benzene-1,4-diol
p-cresol	4-methylphenol

(NCERT Ed. 2025-26 11th Part-II Page No. 262)

78. (4)

$$W = -nR \Delta T = -2 \times 8.314 \times 10 = -166.28 \text{ J}$$

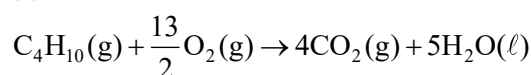
(NCERT Ed. 2025-26 11th Part-I Page No. 142)

79. (3)

- Malonic acid: $\text{HOOC}-\text{CH}_2-\text{COOH}$
- Glutaric acid: $\text{HOOC}-(\text{CH}_2)_3-\text{COOH}$
- Adipic acid: $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$
- Succinic acid: $\text{HOOC}-(\text{CH}_2)_2-\text{COOH}$

(NCERT Ed. 2025-26 11th Part-II Page No. 262)

80. (4)



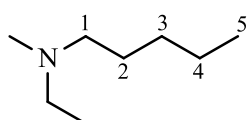
$$\Delta H = \sum(\Delta H_f)_P - \sum(\Delta H_f)_R$$

$$= [4(\Delta_f H)_{\text{CO}_2} + 5(\Delta_f H)_{\text{H}_2\text{O}}] - [(\Delta_f H)_{\text{C}_4\text{H}_{10}}]$$

$$= 4y + 5z - x \text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 150)

81. (1)



N-ethyl-N-methylpentan-1-amine

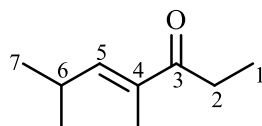
(NCERT Ed. 2025-26 11th Part-II Page No. 267)

82. (1)

- It is impossible to determine lattice enthalpies of ionic compounds directly by experiment.
- For chemical reactions, heat absorbed at constant volume, is measured in a bomb calorimeter.

(NCERT Ed. 2025-26 11th Part-I Page No. 145)

83. (1)



4,6-Dimethylhept-4-en-3-one

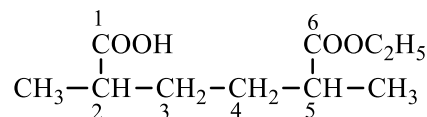
(NCERT Ed. 2025-26 11th Part-II Page No. 267)

84. (2)

$\Delta H < 0$ and $\Delta S < 0$	$\Delta G > 0$ at high temperature
$\Delta H > 0$ and $\Delta S > 0$	$\Delta G > 0$ at low temperatures
$\Delta H < 0$ and $\Delta S > 0$	$\Delta G < 0$ at all temperature
$\Delta H > 0$ and $\Delta S < 0$	$\Delta G > 0$ at all temperatures

(NCERT Ed. 2025-26 11th Part-I Page No. 162)

85. (1)



5-Ethoxycarbonyl-2-methylhexanoic acid

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

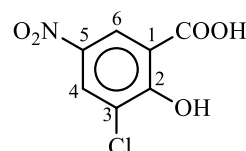
86. (1)

For an irreversible isothermal process:

$$\Delta S_{\text{total}} \neq 0 \text{ and } \Delta U = 0$$

(NCERT Ed. 2025-26 11th Part-I Page No. 159)

87. (2)



3-Chloro-2-hydroxy-5-nitrobenzoic acid

(NCERT Ed. 2025-26 11th Part-II Page No. 269)

88. (2)

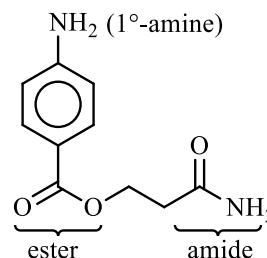
$$q + w = \Delta U$$

$$0 + (-15000) = 10 \times 25 \times (T_2 - 300)$$

$$T_2 = 240 \text{ K}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 140)

89. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 267)

90. (3)

The solubility of most salts in water increases with rise of temperature because for most of ionic compounds, $\Delta_{\text{sol}} H^\circ$ is positive and the dissolution process is endothermic.

(NCERT Ed. 2025-26 11th Part-I Page No. 156)

91. (2)

The first word in a biological name represents the genus while the second component denotes the specific epithet. The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter. Biological names are generally in Latin and written in italics. The correctly written scientific name of wheat is *Triticum aestivum*.

(NCERT Ed. 2025-26, 11th, Page No. 8)

92. (4)

Viroid: It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight.

(NCERT Ed. 2025-26, 11th, Page No. 20-21)

93. (3)

Ernst Mayr was awarded the three prizes widely regarded as the *triple crown* of biology: the *Balzan Prize* in 1983, the *International Prize for Biology* in 1994, and the *Crafoord Prize* in 1999.

(NCERT Ed. 2025-26, 11th, Page No. 2)

94. (4)

In the five kingdom classification, bacteria are included in Kingdom Monera. Diatoms are the chief 'producers' in the oceans. The walls are embedded with silica and thus the walls are indestructible. Very often, red dinoflagellates (Example: *Gonyaulax*) undergo such rapid multiplication that they make the sea appear red (red tides). Slime moulds are saprophytic protists.

(NCERT Ed. 2025-26, 11th, Page No. 5-6)

95. (2)

Man - Family Hominidae
Housefly - Family Muscidae
Mango - Family Anacardiaceae
Petunia - Family Solanaceae

(NCERT Ed. 2025-26, 11th, Page No. 8)

96. (3)

The patients lying in coma in hospitals virtually supported by machines which replace heart and lungs. The patient is otherwise brain-dead. The patient has no self-consciousness.

(NCERT Ed. 2022-23, 11th, Page No. 5)

97. (3)

All single-celled eukaryotes are placed under Protista, but the boundaries of this kingdom are not well defined. What may be 'a photosynthetic protistan' to one biologist may be 'a plant' to another. All protozoans are heterotrophs and live as predators or parasites. They are believed to be primitive relatives of animals.

(NCERT Ed. 2025-26, 11th, Page No. 5)

98. (2)

Genus comprises a group of related species which has more characters in common in comparison to species of other genera. They are separated into two different families – Felidae and Canidae, respectively.

(NCERT Ed. 2025-26, 11th, Page No. 8)

99. (2)

Methanogens are present in the gut of several ruminant animals such as cows and buffaloes and they are responsible for the production of methane (biogas) from the dung of these animals. They are classified as archaebacteria. Archaebacteria differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions. Being prokaryotic, they lack membrane-bound nucleus. Some cyanobacteria can fix atmospheric nitrogen.

(NCERT Ed. 2025-26, 11th, Page No. 4)

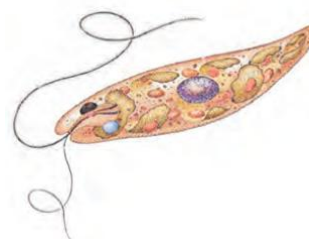
100. (3)

Housefly belongs to order Diptera, Gibbon belongs to order Primata.

(NCERT Ed. 2025-26, 11th, Page No. 8)

101. (3)

The organism shown in the diagram is *Euglena*. It is a unicellular eukaryote. Though they are photosynthetic in the presence of sunlight, when deprived of sunlight they behave like heterotrophs by predating on other smaller organisms.



Euglena

(NCERT Ed. 2025-26, 11th, Page No. 15)

102. (1)

'B' represents the rank of Order. The hierarchy is Class → Order → Family. This means a Class contains one or more Orders, and an Order contains one or more Families. Therefore, an Order is included within a Class. 'C' represents Family. The genus *Datura* belongs to the family Solanaceae. The family Solanaceae is in the order Polymoniales, not Poales. 'D' represents Species. *Mangifera* is the genus for mangoes, and *Felis* is the genus for cats. These organisms belong to different Kingdoms (Plantae and Animalia, respectively) and are not even in the same genus, so can't be in the same species.

(NCERT Ed. 2025-26, 11th, Page No. 8)

103. (3)

Amoeboid protozoans: Marine forms have silica shells on their surface

Flagellated protozoans: The members of this group are either free-living or parasitic. They have flagella. The parasitic forms cause diseases such as sleeping sickness. Example: *Trypanosoma*.

Ciliated protozoans: They have a cavity (gullet) that opens to the outside of the cell surface.

Sporozoans: The most notorious is *Plasmodium* (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.

(NCERT Ed. 2025-26, 11th, Page No. 16)

104. (3)

When a fungus reproduces sexually, two haploid hyphae of compatible mating types come together and fuse. In some fungi the fusion of two haploid cells immediately results in diploid cells (2n). However, in other fungi (ascomycetes and basidiomycetes), an intervening dikaryotic stage (n + n, i.e., two nuclei per cell) occurs; such a condition is called a dikaryon and the phase is called dikaryophase of fungus. Later, the parental nuclei fuse and the cells become diploid.

(NCERT Ed. 2025-26, 11th, Page No. 17)

105. (2)

Most fungi are heterotrophic and absorb soluble organic matter from dead substrates.

(NCERT Ed. 2025-26, 11th, Page No. 16)

106. (2)

In early days, human beings needed to find sources for their basic needs of food, clothing and shelter.

Hence, the earliest classifications were based on the 'uses' of various organisms.

(NCERT Ed. 2025-26, 11th, Page No. 5)

107. (1)

A few members of kingdom Plantae are partially heterotrophic such as the insectivorous plants or parasites. Bladderwort and Venus fly trap are examples of insectivorous plants and *Cuscuta* is a parasite.

(NCERT Ed. 2025-26, 11th, Page No. 19)

108. (1)

Consciousness therefore, becomes the defining property of living organisms.

Human being is the only organism who is aware of himself, i.e., has self-consciousness.

(NCERT Ed. 2022-23, 11th, Page No. 5)

109. (2)

Tetanus is caused by a bacterium (kingdom Monera). *Rhizopus* is the bread mould (kingdom Fungi)

(NCERT Ed. 2025-26, 11th, Page No. 11)

110. (3)

That all present day living organisms are related to each other and also to all organisms that ever lived on this earth, was a revelation which humbled man and led to cultural movements for conservation of biodiversity.

(NCERT Ed. 2025-26, 11th, Page No. 1)

111. (1)

The mycelium is aseptate and coenocytic in *Mucor*, and septate and branched hyphae are present in *Penicillium*. *Anabaena* is a cyanobacterium that possesses specialised cells called heterocysts for fixing atmospheric nitrogen. In prions, the infectious material is abnormally folded protein.

(NCERT Ed. 2025-26, 11th, Page No. 17)

112. (2)

M.W. Beijerinck (1898) demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants. W.M. Stanley (1935) showed that viruses could be crystallised and crystals consist largely of proteins. Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses.

(NCERT Ed. 2025-26, 11th, Page No. 20-21)

- 113. (3)**
Instead of a cell wall, the euglenoids have a protein rich layer called pellicle which makes their body flexible.
(NCERT Ed. 2025-26, 11th, Page No. 15)
- 114. (4)**
Cholera, typhoid, tetanus, citrus canker are well known diseases caused by different bacteria.
(NCERT Ed. 2025-26, 11th, Page No. 21)
- 115. (2)**
The sexual cycle involves the following three steps:
(i) Fusion of protoplasts between two motile or non-motile gametes called plasmogamy.
(ii) Fusion of two nuclei called karyogamy.
(iii) Meiosis in zygote resulting in haploid spores.
(NCERT Ed. 2025-26, 11th, Page No. 17)
- 116. (2)**
Asexual reproduction in fungi is by spores called conidia or sporangiospores or zoospores, and sexual reproduction is by oospores, ascospores and basidiospores. In members of phycomycetes, asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile). A zygospore is formed by fusion of two gametes.
(NCERT Ed. 2025-26, 11th, Page No. 17)
- 117. (1)**
Classification is not a single step process but involves hierarchy of steps in which each step represents a rank or category.
(NCERT Ed. 2025-26, 11th, Page No. 6)
- 118. (2)**
One must remember that increase in body mass is considered as growth. Non-living objects also grow if we take increase in body mass as a criterion for growth. Mountains, boulders and sand mounds do grow. However, this kind of growth exhibited by non-living objects is by accumulation of material on the surface. In living organisms, growth is from inside. Growth, therefore, cannot be taken as a defining property of living organisms. There are many organisms which do not reproduce (mules, sterile worker bees, infertile human couples, etc). Hence, reproduction also cannot be an all-inclusive defining characteristic of living organisms. No non-living object exhibits metabolism.
(NCERT Ed. 2022-23, 11th, Page No. 2-3)
- 119. (2)**
Systematics takes into account evolutionary relationships between organisms. The word systematics is derived from the Latin word 'systema' which means systematic arrangement of organisms.
(NCERT Ed. 2025-26, 11th, Page No. 5-6)
- 120. (4)**
Many members of ascomycetes like morels and truffles are edible and are considered delicacies.
(NCERT Ed. 2025-26, 11th, Page No. 18)
- 121. (2)**
Homo is the genus name, while *sapiens* is the specific epithet. Both Mammalia (mammals) and Insecta (insects) are taxonomic classes. Angiospermae is a major division or phylum within the plant kingdom. Dicotyledonae (dicots) is a class within the angiosperms. Carnivora is an order of mammals. Canidae, which includes dogs, is a family within the order Carnivora.
(NCERT Ed. 2025-26, 11th, Page No. 7-8)
- 122. (1)**
The main criteria for classification used by Whittaker include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms.
(NCERT Ed. 2025-26, 11th, Page No. 11-12)
- 123. (3)**
Humans and dogs belong to kingdom Animalia, phylum Chordata, Class mammalia. Humans belong to order Primata while dog belongs to order Carnivora.
(NCERT Ed. 2025-26, 11th, Page No. 8)
- 124. (1)**
Archaeobacteria are special since they live in some of the most harsh habitats such as extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens).
(NCERT Ed. 2025-26, 11th, Page No. 4)

125. (2)

The Mycoplasma are organisms that completely lack a cell wall. Desmids belong to the group Chrysophytes, commonly known as golden algae. The common mushroom you eat and toadstools are also fungi. In the five kingdom classification of Whittaker there is no mention of lichens and some acellular organisms like viruses, viroids and prions.

(NCERT Ed. 2025-26, 11th, Page No. 14, 21)

126. (3)

Arranging the given ranks in hierarchical order from highest to lowest:

Kingdom (Animalia)

Phylum (Arthropoda)

Class (Monocotyledonae)

Family (Muscidae)

(NCERT Ed. 2025-26, 11th, Page No. 8)

127. (2)

Lichens are symbiotic associations i.e. mutually useful associations, between algae and fungi. The algal component is known as phycobiont and fungal component as mycobiont, which are autotrophic and heterotrophic, respectively. Algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner.

(NCERT Ed. 2025-26, 11th, Page No. 21)

128. (4)

The tobacco mosaic virus infects tobacco cells. It consists of single stranded RNA enclosed within a capsid. All viruses are inert outside specific host cells. The bacteriophage consists of head, collar, sheath and tail fibres.

(NCERT Ed. 2025-26, 11th, Page No. 20-21)

129. (3)

In *Planaria* (flat worms), we observe true regeneration, i.e., a fragmented organism regenerates the lost part of its body and becomes, a new organism.

(NCERT Ed. 2022-23, 11th, Page No. 2)

130. (1)

The diagram in the question represents *Aspergillus*, a member of ascomycetes. The asexual spores are conidia produced exogenously on the special mycelium called conidiophores. Conidia on germination produce mycelium. Sexual spores are

called ascospores which are produced endogenously in sac like asci (singular ascus).

(NCERT Ed. 2025-26, 11th, Page No. 17)

131. (1)

The number of species that are known and described range between 1.7-1.8 million. This refers to biodiversity or the number and types of organisms present on earth. Biologists follow universally accepted principles to provide scientific names to known organisms. Each name has two components – the generic name and the specific epithet. The taxonomic categories from species to kingdom are the seven broad categories. Traditionally, the Kingdom Fungi is divided into four main classes based on The morphology of the mycelium, mode of spore formation and fruiting bodies. These are typically listed as Phycomycetes, Ascomycetes, Basidiomycetes, and Deuteromycetes.

(NCERT Ed. 2025-26, 11th, Page No. 3)

132. (4)

The kingdoms defined by Whittaker were named Monera, Protista, Fungi, Plantae and Animalia. Monera has prokaryotic organisms, while Protista, Fungi, Plantae and Animalia have eukaryotic organisms.

(NCERT Ed. 2025-26, 11th, Page No. 11)

133. (4)

M.W. Beijerinck (1898) demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants and named the new pathogen “virus” and called the fluid as *Contagium vivum fluidum* (infectious living fluid).

(NCERT Ed. 2025-26, 11th, Page No. 20)

134. (2)

In the five kingdom classification of Whittaker, there is no mention of lichens and some acellular organisms like viruses, viroids and prions. Aristotle divided animals into those with red blood and those without it.

(NCERT Ed. 2025-26, 11th, Page No. 1)

135. (3)

Lichens are very good pollution indicators – they do not grow in polluted areas.

(NCERT Ed. 2025-26, 11th, Page No. 21)

136. (1)

- Striations (alternating light and dark bands) are a result of the highly organized, parallel arrangement of the myofilaments (actin and myosin) into functional units called sarcomeres in skeletal and cardiac muscle.
 - Actin and myosin are the fundamental contractile proteins essential for all types of muscle contraction (skeletal, cardiac, and smooth).
 - Smooth muscles lack these well-defined sarcomeres, resulting in a random, less organized arrangement of myofilaments.
 - The lack of this specific, organized pattern is why they are called "non-striated" or "smooth."
- (NCERT Ed. 2025-26, 11th, Page No. 221)

137. (2)

- The urea cycle (Ornithine cycle) occurs in the liver, converting ammonia into urea for safer transport and excretion.
- The dialysing fluid, or dialysate, has the same composition as normal plasma (including Na^+ , K^+ , glucose, etc.) but lacks urea and other nitrogenous wastes, ensuring that these wastes diffuse out of the patient's blood due to the concentration gradient, while essential substances are retained.
- Glomerular filtration is largely non-selective (except for proteins and other large molecules). It prevents the filtration of molecules larger than approximately 69 kDa (size of albumin).
- ANF is released in response to an increase in blood volume/pressure. It causes vasodilation (not vasoconstriction) and acts as a check on the RAAS mechanism. This leads to an overall decrease in blood pressure and GFR regulation towards normal (natriuresis).

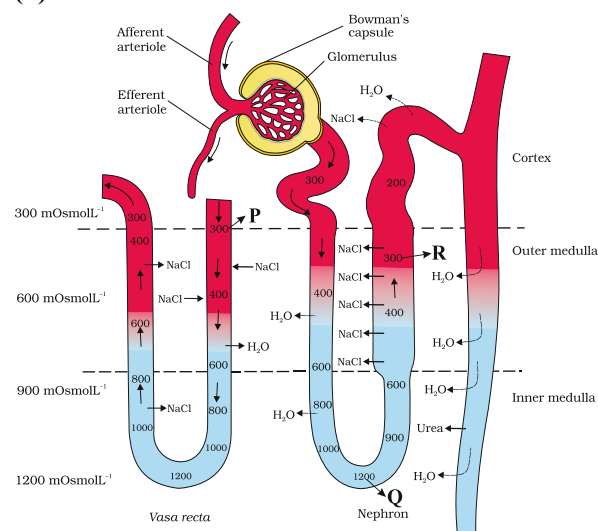
(NCERT Ed. 2025-26, 11th, Page No. 212)

138. (3)

- A myofibril is made of repeating sarcomeres containing actin (thin) and myosin (thick) filaments; many myofibrils pack together to form a muscle fiber (muscle cell).
- Red (slow-twitch/oxidative) fibers are rich in myoglobin and mitochondria and use aerobic respiration; white (fast-twitch/glycolytic) fibers have less myoglobin and rely more on anaerobic glycolysis.

(NCERT Ed. 2025-26, 11th, Page No. 226)

139. (1)



The proximity between the Henle's loop and *vasa recta*, as well as the counter current in them help in maintaining an increasing osmolarity towards the inner medullary interstitium, i.e., from 300 mOsmolL^{-1} in the cortex to about 1200 mOsmolL^{-1} in the inner medulla. The ascending limb is impermeable to water but allows transport of electrolytes actively or passively. Therefore, as the concentrated filtrate pass upward, it gets diluted (300 mOsmolL^{-1}) due to the passage of electrolytes to the medullary fluid.

(NCERT Ed. 2025-26, 11th, Page No. 211)

140. (2)

List-I (Bone)	List-II (Body Part)
Scapula	Shoulder Girdle
Tarsals	Ankle/Foot
Tibia	Lower Leg
Radius	Forearm

(NCERT Ed. 2025-26, 11th, Page No. 228)

141. (3)

The active reabsorption of NaCl from the thick ascending limb of the Loop of Henle is the single most crucial step in establishing the high osmotic gradient (counter-current multiplier mechanism) in the medullary interstitium. If this active transport is completely inhibited, the osmotic gradient would largely wash out. Without a strong gradient, the maximum osmolarity would drop significantly from 1200 mOsmol/L , and the body would be unable to produce concentrated urine effectively.

(NCERT Ed. 2025-26, 11th, Page No. 211)

142. (2)

Cardiac muscle fibres are branched, striated, involuntary, and joined by intercalated discs (containing gap junctions and desmosomes).

Tetany is rapid spasms (wild contractions) in muscle due to low Ca^{++} in body fluid.

The A-band is the dark band. It represents the full length of the thick (myosin) filaments and includes the regions where thin (actin) filaments overlap them.

Synovial joints are lubricated by synovial fluid, which is secreted by the synovial membrane lining the joint capsule, not the articular cartilage.

Fibrous joints do not allow any movement. This type of joint is shown by the flat skull bones which fuse end-to-end with the help of dense fibrous connective tissues in the form of sutures, to form the cranium.

(NCERT Ed. 2025-26, 11th, Page No. 221)

143. (2)

The vertebral column is differentiated into cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused) regions starting from the skull.

(NCERT Ed. 2025-26, 11th, Page No. 227)

144. (1)

List I (Excretory Product)	List II (Animal Group)
Excretory organ in <i>Amphioxus</i>	Flame cells
Human kidney	10-12 cm in length
Ammonotelic	Bony Fishes
Uricotelic	Crocodiles

(NCERT Ed. 2025-26, 11th, Page No. 206)

145. (1)

- The Myosin head has a binding site for ATP and acts as an ATPase, releasing energy for the muscle contraction cycle.
- The power stroke is the movement that pulls the Actin filament towards the center of the A-band. It occurs when ADP and P_i are released from the Myosin head. The Myosin head detaches after the power stroke, when a new ATP molecule binds, and re-cocks when the ATP is hydrolysed again.

(NCERT Ed. 2025-26, 11th, Page No. 225)

146. (3)

- The counter-current multiplier (Loop of Henle) and the counter-current exchanger (*Vasa recta*) are essential for building and maintaining the high osmotic gradient in the renal medulla, which is crucial for drawing water out of the collecting duct to produce highly concentrated (hypertonic) urine.
- NaCl is actively transported out of the thick ascending limb of the Loop of Henle into the medullary interstitium. Urea recycling occurs when it is reabsorbed from the collecting duct back into the medulla, significantly boosting the interstitial osmolarity in the deepest parts of the kidney.
- Facultative water reabsorption refers to water movement that is regulated by hormones, specifically Antidiuretic Hormone (ADH) or vasopressin. ADH increases the permeability of the DCT and CD to water, allowing reabsorption to occur based on the body's hydration needs. The early parts of the nephron (PCT) perform obligatory water reabsorption, which is unregulated.
- Ureotelic organisms (like mammals) convert toxic ammonia to the less toxic urea, which can be excreted with less water than ammonia. Ammonotelic organisms (like most aquatic animals) excrete highly toxic ammonia directly, which requires large volumes of water for safe dilution and removal.
- Ammonia is highly neurotoxic and requires high dilution to prevent systemic harm. Urea is approximately 100,000 times less toxic than ammonia, allowing it to be safely concentrated and excreted in a smaller volume of water.

(NCERT Ed. 2025-26, 11th, Page No. 205)

147. (2)

Muscle contraction is initiated when Ca^{2+} is released from the sarcoplasmic reticulum. The correct sequence is:

1. Ca^{2+} binds to the TpC subunit of the regulatory protein Troponin.
2. This binding induces a conformational change in the entire Troponin complex.
3. This change pulls the associated filament, Tropomyosin, away from the active sites on the Actin (G-actin) filament.
4. The exposed active sites on Actin allow the Myosin heads to bind, initiating the cross-bridge cycle (sliding filament theory).

(NCERT Ed. 2025-26, 11th, Page No. 224)

148. (3)

The human Pelvic Girdle consists of two Coxal bones (or innominate bones). Each coxal bone is formed by the fusion of three bones: the Ilium, the Ischium, and the Pubis.

(NCERT Ed. 2025-26, 11th, Page No. 229)

149. (4)

Sweat is a watery fluid that contains NaCl, small amounts of urea, and lactic acid, but does not typically contain cholesterol. Cholesterol is primarily managed by the liver.

(NCERT Ed. 2025-26, 11th, Page No. 213)

150. (3)

During muscle contraction, the myosin head forms a strong cross-bridge with the actin filament. For the muscle to relax or continue the sliding process, the cross-bridge must be broken. The crucial step for breaking this strong bond is the binding of a new ATP molecule to the myosin head. If ATP is depleted, the cross-bridges cannot detach, and the muscle fibers become locked in a rigid, contracted state.

(NCERT Ed. 2025-26, 11th, Page No. 225)

151. (3)

Sebaceous glands secrete sebum, which contains sterols, hydrocarbons, and waxes. This secretion provides an oily protective covering for the skin.

(NCERT Ed. 2025-26, 11th, Page No. 213)

152. (2)

The elbow joint, specifically the articulation between the Humerus (trochlea) and the Ulna (trochlear notch), is a classic example of a Hinge Joint. Hinge joints primarily allow movement in a single plane, which is Flexion and Extension (bending and straightening the arm).

(NCERT Ed. 2025-26, 11th, Page No. 231)

153. (3)

- Human skull has 8 Cranial + 14 Facial = 22 bones (excluding hyoid and ear ossicles).
- Girdles (Pectoral and Pelvic) are the connecting parts between the axial skeleton and the limbs (Appendicular skeleton).
- Synovial joints can be uniaxial (e.g., Hinge/Pivot), biaxial (e.g., Condylar/Saddle), or multiaxial (e.g., Ball and socket). Not all are biaxial.

- Chondrocytes are the cells of cartilage. The cells responsible for synthesizing and secreting the organic matrix of bone are osteoblasts.

(NCERT Ed. 2025-26, 11th, Page No. 227)

154. (2)

The Glomerular Filtration Membrane consists of three main layers, which the fluid passes through in this order:

1. Endothelium of the Glomerular Capillary: Contains pores called fenestrations.
2. Basement Membrane: A thin layer of glycoprotein and collagen.
3. Epithelium of Bowman's Capsule (Podocytes): The podocytes have filtration slits (slit pores) between their pedicels.

(NCERT Ed. 2025-26, 11th, Page No. 208)

155. (4)

Limb bones, such as the long bones of the legs, serve weight-bearing functions.

Function of ciliated epithelium is to move particles or mucus in a specific direction over the epithelium.

In all connective tissues except blood, the cells secrete fibres.

Cell junctions fuse the plasma membranes of cardiac muscle cells and make them stick together.

(NCERT Ed. 2025-26, 11th, Page No. 61)

156. (3)

Presence of glucose (Glycosuria) and ketone bodies (Ketonuria) in urine are indicative of diabetes mellitus.

In diabetes insipidus, the volume of urine output increases dramatically (polyuria). This is due to the inefficiency or lack of production of ADH (Vasopressin), not its efficiency. Without ADH, the collecting duct cannot reabsorb water, leading to excessive water loss.

If the CD receptors are faulty or unresponsive to ADH (even if ADH levels are normal or high), the CD remains impermeable to water. The dilute filtrate from the Loop of Henle cannot be concentrated, leading to the excretion of dilute urine. Collecting Duct allows passage of small amounts of urea into the medullary interstitium to keep up the osmolarity.

(NCERT Ed. 2025-26, 11th, Page No. 212)

157. (4)

Urea is reabsorbed passively PCT and CD and recycled ascending limb of LOH. It is not actively secreted into the PCT.

Nearly all of the essential nutrients, and 70-80 per cent of electrolytes and water are reabsorbed by this segment.

Glucose reabsorption is mainly in the PCT via secondary active transport (co-transport with Na^+), followed by facilitated diffusion on the basolateral side.

Nearly 80-90% of filtered HCO_3^- is reabsorbed in the PCT. It is not reabsorbed directly; instead, it combines with secreted H^+ to form H_2CO_3 , which is broken down by carbonic anhydrase (on the brush border) into CO_2 and H_2O . CO_2 then diffuses back into the cell, and the process regenerates new HCO_3^- in the cell, which then moves into the blood.

(NCERT Ed. 2025-26, 11th, Page No. 209)

158. (3)

- The filtration process is determined by Starling forces. Due to the high permeability of the glomerular capillary wall, all small solutes (electrolytes, glucose, amino acids, urea, etc.) pass freely into Bowman's capsule, achieving nearly the same concentration as in the plasma.
- The absence of plasma proteins in the filtrate creates an osmotic force (Colloid Osmotic Pressure) that opposes filtration, but the net effect of all solutes being equalized ensures that the total osmolarity of the filtrate in the Bowman's capsule ($\approx 300\text{mOsmol L}^{-1}$) is almost the same as the plasma.

(NCERT Ed. 2025-26, 11th, Page No. 208)

159. (1)

The H-zone is the central region of the A-band where the thin (Actin) filaments do not overlap with the thick (Myosin) filaments in a relaxed muscle. Therefore, a cross-section through the H-zone would only contain the thick (Myosin) filaments.

(NCERT Ed. 2025-26, 11th, Page No. 223)

160. (4)

The micturition reflex is fundamentally a neural reflex triggered by the activation of stretch receptors in the bladder wall as it fills. This reflex is coordinated by the CNS. However, in adults, the

reflex is modulated by higher brain centers, allowing for voluntary control over the external urethral sphincter (which is skeletal muscle), thus enabling the conscious delay or initiation of urination.

(NCERT Ed. 2025-26, 11th, Page No. 212)

161. (4)

A sarcomere is the segment between two successive Z-lines. The structure from Z-line to Z-line is: Z-line - I-band/2A-band (A-band contains H-zone, which contains the M-line) - I-band/2 - Z-line.

Travelling from the centre of one sarcomere (which is the M-line) to the centre of an adjacent sarcomere (its M-line):

1. Start: M-line (Centre of sarcomere 1)
2. Move Out: H-zone (Thick filaments only)
3. Move Further Out: A-band (Thick and Thin overlap)
4. Move to End of Sarcomere 1: I-band/2 (Thin filaments only)
5. Boundary: Z-line (End of sarcomere 1/Start of sarcomere 2)
6. Move into Sarcomere 2: I-band/2 (Thin filaments only)
7. Move Further In: A-band (Thick and Thin overlap)
8. Near Centre: H-zone (Thick filaments only)
9. End: M-line (Centre of sarcomere 2)

Sequence: M H A I Z I A H M.

(NCERT Ed. 2025-26, 11th, Page No. 223)

162. (2)

- The duct that carries urine from the renal pelvis to the urinary bladder is the ureter.
- The DCT (and collecting duct) are crucial for conditional reabsorption and secretion. The secretion of H^+ , K^+ and ammonia into the filtrate, along with selective reabsorption of H_2CO_3 , primarily occurs here to regulate the pH and ionic balance of the blood. While some secretion also occurs in PCT, the DCT is the site of K^+ secretion regulated by aldosterone and is primarily associated with fine-tuning the balance.

(NCERT Ed. 2025-26, 11th, Page No. 209)

163. (4)

Approximately 80-85% of all nephrons are cortical nephrons. They are located primarily in the renal cortex. Juxtamedullary nephrons make up the remaining 15-20%.

The *vasa recta* (the capillary network surrounding the Loop of Henle) is specifically associated with the long loops of Henle found in juxtamedullary nephrons. In cortical nephrons, the loop of Henle is short.

When the body is acidic, the kidneys compensate by:

1. Secreting H^+ (hydrogen ions) into the filtrate, removing acid from the body.
2. Reabsorbing HCO_3^- (bicarbonate ions) back into the blood, where it acts as a buffer.

Cortical nephrons have a short loop of Henle that primarily remains in the cortex but extends slightly into the outer medulla, not rarely. The statement should say that it dips slightly into the medulla, not rarely does so. The juxtamedullary nephrons are the ones with the long loops that extend deep into the inner medulla.

The renal artery delivers blood to the kidney for filtration. The renal vein carries filtered blood away from the kidney. Since the kidney's function is to remove urea and other wastes from the blood and excrete them in the urine, the concentration of urea is actually highest in the renal artery (before filtration) and lowest in the renal vein (after filtration).

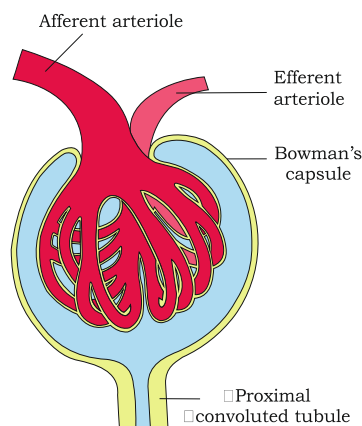
(NCERT Ed. 2025-26, 11th, Page No. 208)

164. (2)

List-I	List-II
<i>Paramoecium</i>	Cilia helps in movement of food through cytopharynx
<i>Hydra</i>	Uses tentacles for capturing its prey
<i>Vasa recta</i>	Counter-current exchanger
Malpighian Tubules	Helps in excretion in insects

(NCERT Ed. 2025-26, 11th, Page No. 206)

165. (4)



(NCERT Ed. 2025-26, 11th, Page No. 207)

166. (3)

- Dialysis is a clinical procedure that attempts to mimic the kidney's clearance function.
- The patient's blood is passed through a dialyzing unit. The unit contains a semi-permeable membrane separating the blood from the dialyzing fluid (dialysate).
- The dialysate is formulated to have the desired concentration of essential electrolytes (like Na^+) and no waste products (like urea).
- Waste products (e.g., Urea, Creatinine) move out of the blood into the dialysate down their concentration gradient via Diffusion.
- Excess fluid is also removed by creating a pressure difference across the membrane, which is a form of Ultrafiltration.

(NCERT Ed. 2025-26, 11th, Page No. 214)

167. (1)

Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called fascia.

- Excitability: Ability to respond to stimuli.
- Contractility: Ability to shorten and thicken (contract).
- Extensibility: Ability to stretch beyond resting length.
- Elasticity: The ability to recoil and return to its original length after being stretched or contracted. This is the correct property.

(NCERT Ed. 2025-26, 11th, Page No. 222)

168. (2)

ANP's actions:

- It inhibits the secretion of renin from the juxtaglomerular cells in the kidneys.
- This, in turn, inhibits the formation of angiotensin II and the release of aldosterone from the adrenal gland.
- It causes vasodilation of blood vessels, particularly the afferent arterioles in the kidneys, and constriction of the efferent arterioles, which increases the glomerular filtration rate (GFR).
- It inhibits sodium and water reabsorption in the renal tubules (especially the collecting ducts), leading to increased sodium and water excretion (natriuresis and diuresis).
- These combined effects lead to a decrease in blood volume and blood pressure.

(NCERT Ed. 2025-26, 11th, Page No. 212)

169. (4)

Amoeboid movement is carried out by pseudopodia formed by the streaming of protoplasm, a process regulated by the cytoskeletal microfilaments. This type of movement is characteristic of macrophages and leucocytes (White Blood Cells) in the blood, allowing them to squeeze through capillary walls.

(NCERT Ed. 2025-26, 11th, Page No. 221)

170. (2)

- Protonephridia or Flame Cells are the excretory structures in Rotifers, Platyhelminthes (Planaria), and some Annelids. (Correct)
- Prawns (Crustaceans) possess Antennal Glands (or Green Glands) as their excretory organs. Nephridia are the tubular excretory structures typically found in Earthworms and other Annelids. (Incorrect Pair)
- Antennal Glands or Green Glands are the excretory structures in Crustaceans like Prawns. (Correct)
- Flame Cells are the alternative name for Protonephridia and are the excretory structures in Platyhelminthes.

(NCERT Ed. 2025-26, 11th, Page No. 206)

171. (2)

List-I	List-II
Myasthenia Gravis	Autoimmune disorder affecting neuromuscular junction
Muscular Dystrophy	Progressive degeneration of skeletal muscle
Osteoporosis	Decreased level of oestrogen is a common cause
Gout	Accumulation of uric acid crystals in joints

(NCERT Ed. 2025-26, 11th, Page No. 231)

172. (3)

ACE is an enzyme that catalyzes the conversion of Angiotensin I to the potent vasoconstrictor and Aldosterone-releasing agent, Angiotensin II.

- Inhibiting ACE stops the formation of Angiotensin II.
- Since Angiotensin II is the primary stimulus for the release of Aldosterone from the adrenal cortex, the drug will lead to reduced Aldosterone levels.
- Reduced Aldosterone leads to less Na^+ reabsorption and thus increased Na^+ excretion (Natriuresis) and accompanying water loss, which results in a decrease in blood volume and blood pressure BP. ACE inhibitors are a common class of drugs for treating hypertension.

(NCERT Ed. 2025-26, 11th, Page No. 212)

173. (1)

- This unit called the sarcomere, which is the fundamental contractile unit of striated muscle is demarcated by two successive Z-lines (or Z-discs).
- Each vertebra has a central hollow portion (neural canal) through which the spinal cord passes.
- First vertebra is the atlas and it articulates with the occipital condyles.
- Bones store the majority of the body's calcium and phosphate. Iron is primarily stored in the liver, spleen, and bone marrow (as part of hemoglobin/ferritin).
- Locomotion is the movement of an organism from one place to another (e.g., walking). Movement is any change in position of a body part (e.g., blinking). Therefore, locomotion is a subset of movement.

(NCERT Ed. 2025-26, 11th, Page No. 227)

174. (2)

The $\text{Na}^+\text{-K}^+$ pump is located on the basolateral membrane of the PCT cells. Its function is to actively pump Na^+ out of the cell into the interstitial fluid.

- This action maintains a low intracellular Na^+ concentration.
- The resulting steep Na^+ concentration gradient across the apical membrane provides the driving force for the secondary active transport (co-transport) of substances like glucose and amino acids (via Na^+ Glucose co-transporter, etc.) from the filtrate into the cell.
- A failure of the pump eliminates this Na^+ gradient, thus abolishing the reabsorption of glucose, amino acids, and bicarbonate.

(NCERT Ed. 2025-26, 11th, Page No. 209)

175. (1)

List-I	List-II
Pivot Joint	Between Atlas and Axis
Gliding Joint	Between Carpals
Saddle Joint	Between Carpal and Metacarpal of Thumb
Ball and Socket Joint	Between Humerus and Pectoral Girdle

(NCERT Ed. 2025-26, 11th, Page No. 231)

176. (1)

Synovial joints are characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones. Such an arrangement allows considerable movement. Ball and socket joint (between humerus and pectoral girdle), hinge joint (knee joint), pivot joint (between atlas and axis), gliding joint (between the carpals) and saddle joint (between carpal and metacarpal of thumb) are some examples.

(NCERT Ed. 2025-26, 11th, Page No. 230)

177. (2)

JG cells are indeed modified smooth muscle cells in the wall of the afferent arteriole. They sense a drop in blood pressure (BP) and respond by secreting Renin, initiating the Renin-Angiotensin-Aldosterone System (RAAS) to restore BP.

The kidneys regulate the plasma concentrations of critical electrolytes (like Na^+ , K^+ , Ca^{2+} , Cl^-) and are the main long-term regulators of acid-base balance by controlling the excretion of H^+ and reabsorption of HCO_3^- (bicarbonate).

The patella (kneecap) is the largest example of a sesamoid bone, which is a bone embedded within a tendon (in this case, the tendon of the quadriceps femoris muscle).

Towards the centre of the inner concave surface of the kidney is a notch called hilum is present through which ureter, blood vessels and nerves enter.

The sarcolemma is the specialized plasma membrane that encloses the cytoplasm (sarcoplasm) of a muscle cell or muscle fibre.

(NCERT Ed. 2025-26, 11th, Page No. 209)

178. (1)

- A motor neuron along with the muscle fibres connected to it constitute a motor unit.
- The A-band is defined by the length of the thick myosin filaments. Muscle contraction is due to the sliding of the thin filaments past the thick filaments, not a shortening of the filaments themselves. Therefore, the A-band length remains constant, and the Reason (R) is the precise explanation for the constant width of the A-band.

(NCERT Ed. 2025-26, 11th, Page No. 224)

179. (1)

When GFR drops (e.g., due to low blood pressure), the JGA initiates the RAAS cascade. Angiotensin II, formed from Angiotensin I, is a powerful vasoconstrictor. Its selective, potent action on the efferent arteriole increases the resistance to blood flow leaving the glomerulus. This 'damming' effect raises the Glomerular Hydrostatic Pressure (GHP) within the glomerulus, thereby increasing Net Filtration Pressure (NFP) and helping to restore the GFR towards normal.

(NCERT Ed. 2025-26, 11th, Page No. 212)

180. (3)

A single U-shaped bone called hyoid is present at the base of the buccal cavity. This is not attached to any other bone.

(NCERT Ed. 2025-26, 11th, Page No. 227)

■■■

